

## Packet Tracer Lab 8 - Connect a Router to a LAN

### Addressing Table

| Device | Interface    | IP Address      | Subnet Mask     | Default Gateway |
|--------|--------------|-----------------|-----------------|-----------------|
| R1     | G0/0         | 192.168.10.1    | 255.255.255.0   | N/A             |
|        | G0/1         | 192.168.11.1    | 255.255.255.0   | N/A             |
|        | S0/0/0 (DCE) | 209.165.200.225 | 255.255.255.252 | N/A             |
| R2     | G0/0         | 10.1.1.1        | 255.255.255.0   | N/A             |
|        | G0/1         | 10.1.2.1        | 255.255.255.0   | N/A             |
|        | S0/0/0       | 209.165.200.226 | 255.255.255.252 | N/A             |
| PC1    | NIC          | 192.168.10.10   | 255.255.255.0   | 192.168.10.1    |
| PC2    | NIC          | 192.168.11.10   | 255.255.255.0   | 192.168.11.1    |
| PC3    | NIC          | 10.1.1.10       | 255.255.255.0   | 10.1.1.1        |
| PC4    | NIC          | 10.1.2.10       | 255.255.255.0   | 10.1.2.1        |

### Objectives

**Part 1: Display Router Information**

**Part 2: Configure Router Interfaces**

**Part 3: Verify the Configuration**

### Background

In this activity, you will use various **show** commands to display the current state of the router. You will then use the Addressing Table to configure router Ethernet interfaces. Finally, you will use commands to verify and test your configurations.

**Note:** The routers in this activity are partially configured. Some of the configurations are not covered in this course but they are provided to assist you in using verification commands.

### Part 1: Display Router Information

#### Step 1: Display interface information on R1.

**Note:** Click a device and then click the **CLI** tab to access the command line directly. The console password is **cisco**. The privileged EXEC password is **class**.

a. Which command displays the statistics for all interfaces configured on a router? **show interfaces**

b. Which command displays the information about the Serial 0/0/0 interface only? **show interfaces serial 0/0/0**

c. Enter the command to display the statistics for the Serial 0/0/0 interface on R1 and answer the following questions:

1) What is the IP address configured on R1? **Internet address is 209.165.200.225/30**

2) What is the bandwidth on the Serial 0/0/0 interface? **BW 1544 Kbit**

d. Enter the command to display the statistics for the GigabitEthernet 0/0 interface and answer the following questions:

1) What is the IP address on R1? **No IP address listed**

2) What is the MAC address of the GigabitEthernet 0/0 interface? **address is 000d.bd6c.7d01 (bia 000d.bd6c.7d01)**

3) What is the bandwidth (BW) of the GigabitEthernet 0/0 interface? **BW 100000 Kbit**

## Step 2: Display a summary list of the interfaces on R1.

a. Which command displays a brief summary of the current interfaces, interface status, and the IP addresses assigned to them? **show ip interface brief**

b. Enter the command on each router and answer the following questions:

1) How many serial interfaces are there on R1 and R2? **R1=02, R2=02**

2) How many Ethernet interfaces are there on R1 and R2? **R1=06, R2=02**

3) Are all the Ethernet interfaces on R1 the same? **No** If no, explain the difference(s). **There are (2) GigabitEthernet interfaces and (4) FastEthernet interfaces. There are two Gigabit Ethernet interfaces and 4 Fast Ethernet interfaces. Gigabit Ethernet interfaces support speeds of up to 1,000,000,000 bits per second and Fast Ethernet interfaces support speeds of up to 1,000,000 bits per second.**

## Step 3: Display the routing table on R1.

a. What command displays the contents of the routing table? **show ip route**

b. Enter the command on R1 and answer the following questions:

1) How many connected routes are there (uses the C code)? **1**

2) Which route is listed? **209.165.200.224/30 is directly connected, Serial0/0/0**

3) How does a router handle a packet destined for a network that is not listed in the routing table? **The router drops the packet (discarding it) and, in many cases, sends an Internet Control Message Protocol (ICMP) "Destination Unreachable" message back to the sender**

4) What commands can you use to verify: **show run, show interfaces, show ip protocols**

## Part 2: Configure Router Interfaces

### Step 1: Configure the GigabitEthernet 0/0 interface on R1.

- a. Enter the following commands to address and activate the GigabitEthernet 0/0 interface on **R1**:

```
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
```

- b. It is good practice to configure a description for each interface to help document the network. Configure an interface description that indicates the device to which it is connected.

```
R1(config-if)# description LAN connection to S1
```

- c. **R1** should now be able to ping PC1.

```
R1(config-if)# end
%SYS-5-CONFIG_I: Configured from console by console
R1# ping 192.168.10.10
```

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 192.168.10.10, timeout is 2 seconds:
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/8 ms
```

### Step 2: Configure the remaining Gigabit Ethernet Interfaces on R1 and R2.

- a. Use the information in the Addressing Table to finish the interface configurations for **R1** and **R2**. For each interface, do the following:
- 1) Enter the IP address and activate the interface.
  - 2) Configure an appropriate description.
- b. Verify interface configurations.

### Step 3: Back up the configurations to NVRAM.

Save the configuration files on both routers to NVRAM. What command did you use? **Copy running-config startup-config**

## Part 3: Verify the Configuration

### Step 1: Use verification commands to check your interface configurations.

- a. Use the **show ip interface brief** command on both **R1** and **R2** to quickly verify that the interfaces are configured with the correct IP address and are active.

How many interfaces on **R1** and **R2** are configured with IP addresses and in the “up” and “up” state? **03 and 03, 06 total**

What part of the interface configuration is NOT displayed in the command output? **Subnet Mask, Interface Description, Secondary IP Addresses, Layer 2 Details like MAC addresses, Performance Statistics, Routing and Helper Data**

What commands can you use to verify this part of the configuration? **Show running-config**

b. Use the **show ip route** command on both **R1** and **R2** to view the current routing tables and answer the following questions:

Questions

1) How many connected routes (uses the **C** code) do you see on each router? **R1=03, R2=03**

2) How many OSPF routes (uses the **O** code) do you see on each router? **R1=02, R2=02**

3) If the router knows all the routes in the network, then the number of connected routes and dynamically learned routes (OSPF) should equal the total number of LANs and WANs. How many LANs and WANs are in the topology? **5 LANS and WANS**

4) Does this number match the number of C and O routes shown in the routing table? **Yes, 3 Cs, 2 Os**

**Note:** If your answer is “no”, then you are missing a required configuration. Review the steps in Part 2.

## Step 2: Test end-to-end connectivity across the network.

You should now be able to ping from any PC to any other PC on the network. In addition, you should be able to ping the active interfaces on the routers. For example, the following tests should be successful:

- From the command line on PC1, ping PC4.
- From the command line on R2, ping PC2.

**Note:** For simplicity in this activity, the switches are not configured. You will not be able to ping them.